

ITA0609-MACHINE LEARNING

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Question: 1

1. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

Answer:

Given:

Section A = 10 cars

Section B = 15 cars

Section C = 15 cars

Since Section B is twice as likely,

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$$

Probability of selecting a blue car:

$$P(\text{Blue}|A) = \frac{3}{10}, P(\text{Blue}|B) = \frac{6}{15} = \frac{2}{5}, P(\text{Blue}|C) = \frac{2}{15}$$

Step 1: Find $P(\text{Blue})$

$$P(\text{Blue}) = \frac{3}{10} \times \frac{1}{4} + \frac{2}{5} \times \frac{1}{2} + \frac{2}{15} \times \frac{1}{4} = \frac{3}{40} + \frac{1}{5} + \frac{1}{30}$$

Taking LCM = 120,

$$= \frac{9}{120} + \frac{24}{120} + \frac{4}{120} = \frac{37}{120}$$

Step 2: Apply Bayes' Theorem

$$P(C|\text{Blue}) = \frac{P(\text{Blue}|C) \times P(C)}{P(\text{Blue})} = \frac{\frac{2}{15} \times \frac{1}{4}}{\frac{37}{120}} = \frac{4}{37} = 0.108$$

Final Answer:

$$P(C|\text{Blue}) = \frac{4}{37} = 0.108 \text{ (10.8\%)}$$

Question: 2

2. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

1. (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
2. (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
3. (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
4. (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
5. (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
6. (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
7. (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
8. (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2
9. i) Identify the epoch at which the model begins to overfit.
10. ii) Suggest two effective techniques to reduce overfitting in this model.
11. iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Answer:

Step 1: Observe the Training Loss

Training loss decreases continuously from **2.8 to 1.3**.

Step 2: Observe the Validation Loss

Validation loss decreases until **Epoch 3**, then starts increasing from **Epoch 4**.

Step 3: Identify Overfitting

Since training loss decreases while validation loss increases,

Overfitting starts at Epoch 4.

Techniques to Reduce Overfitting

- Early Stopping
- Dropout (or L2 Regularization)

Effect on Generalization

Overfitting causes the model to memorize the training data instead of learning general patterns. As a result, it performs well on training data but poorly on new or unseen data.

i) Overfitting begins at Epoch 4.

ii) Early Stopping and Dropout.

iii) Overfitting reduces the model's generalization ability and lowers performance on unseen data.

Question: 3

3. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

12. (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

13. (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%

14. (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%

15. (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%

16. (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%

17. (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%

18. (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%

19. (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%

20. i) Identify the epoch at which the model starts overfitting.

21. ii) Explain why validation accuracy decreases despite training

accuracy increasing. 22. iii) Suggest two methods to improve validation performance

Answer :

Step 1: Observe the Training Accuracy

Epoch Training Accuracy

1	60%
2	65%
3	70%
4	75%
5	80%
6	85%
7	88%
8	92%

Training accuracy **continuously increases** from **60% to 92%**.

Step 2: Observe the Validation Accuracy

Epoch Validation Accuracy

1	58%
2	63%
3	68%
4	72%
5	73%
6	72%
7	70%
8	68%

Validation accuracy **increases up to Epoch 5 (73%)** and then **decreases from Epoch 6**.

Step 3: Identify Overfitting

- Training accuracy keeps increasing.
- Validation accuracy starts decreasing after Epoch 5.

Therefore,

Overfitting starts at Epoch 6.

Step 4: Explain the Reason

The model begins to **memorize the training data** instead of learning general patterns.

As a result:

- Training accuracy continues to increase.
- Validation accuracy decreases because the model performs poorly on unseen data.

Step 5: Methods to Improve Validation Performance

1. Early Stopping

- Stop training when validation accuracy starts decreasing.

2. Dropout (or L2 Regularization)

- Prevents the model from overfitting by reducing model complexity.

Final Answers

i) The model starts **overfitting at Epoch 6**.

ii) Validation accuracy decreases because the model **overfits the training data** and cannot generalize well to new data.

iii) Two methods to improve validation performance are:

- **Early Stopping**
- **Dropout (or L2 Regularization)**